Project Title

AI-Based CV Creation and Optimization using Large Language Models (LLMs)

Abstract

This project presents an AI-driven system for automated resume creation and optimization using Large Language Models (LLMs). The primary objective is to assist users in generating professional, ATS-friendly resumes tailored to specific job descriptions. The system processes resumes in multiple formats (PDF, DOCX) and extracts structured information such as skills, experience, and education using LLM-based parsing techniques. Similarly, job descriptions are analyzed to extract requirements, responsibilities, and key skills.

A core component of the system is resume tailoring, where the LLM rewrites and enhances candidate information by aligning it with job requirements and incorporating relevant keywords for improved Applicant Tracking System (ATS) compatibility. Additionally, the system provides an iterative feedback mechanism that evaluates resume quality, identifies missing keywords, and suggests improvements.

The final output is a well-structured resume in DOCX format. The system is implemented using Python, Streamlit for UI, and open-source LLMs via Ollama. The results demonstrate effective automation of resume generation and optimization, making the system useful for job seekers aiming to improve their chances in recruitment processes.

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1. Introduction

Context and Background

In modern recruitment systems, Applicant Tracking Systems (ATS) are widely used to filter resumes based on keyword matching and formatting. Many candidates fail to pass these systems due to poorly structured resumes or lack of relevant keywords.

Motivation

The motivation behind this project is to leverage advancements in Large Language Models (LLMs) to automate resume creation, improve content quality, and enhance alignment with job requirements. This reduces manual effort and increases the chances of shortlisting.

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2. Problem Statement

Creating a job-specific, ATS-friendly resume is a time-consuming and skill-intensive process. Candidates often struggle with:

- Identifying relevant keywords from job descriptions
- Structuring resumes professionally
- Tailoring content for different job roles

The problem is to design an automated system that:

- Extracts meaningful information from resumes

- Analyzes job descriptions
- Generates optimized resumes tailored to specific role

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3. Objectives

- Extract structured data from resumes (PDF/DOCX)
- Parse job descriptions into key components
- Generate tailored, ATS-friendly resumes
- Incorporate relevant keywords automatically
- Provide ATS feedback (match score, missing keywords)
- Enable iterative refinement through user interaction

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4. Methodology

Tools and Technologies Used

- **Programming Language:** Python
- **LLM Runtime:** Ollama
- **Model:** Gemma (open-source LLM)
- **Frontend:** Streamlit
- **Libraries:** pdfplumber, python-docx, request

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5. Modules and Functionality

1. Resume Extraction Module

- Converts PDF/DOCX to text
- Uses LLM to extract structured fields:
    - Name, contact, skills, experience, education

2. Job Description Parsing Module

- Accepts text or file input
- Extracts:
    - Requirements
    - Responsibilities
    - Keywords

3. Resume Tailoring Module

- Uses LLM to:
    - Rewrite resume content
    - Align with job requirements
    - Inject relevant keywords
- Ensures ATS-friendly formatting

4. ATS Feedback Module

- Evaluates resume vs job description
- Outputs:
  - Match score
  - Missing keywords
  - Improvement suggestions

5. Iterative Refinement Module

- Allows user edits
- Re-runs LLM for improvement
- Enables continuous optimization

6. Document Generation Module

- Converts structured data into DOCX format
- Applies headings, bullet points, and formatting

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6. Results and Analysis

Key Findings

- The system successfully generates tailored resumes aligned with job descriptions
- Keyword matching significantly improves resume relevance
- Iterative refinement enhances resume quality over multiple cycles
- Structured JSON pipeline improves consistency and reliability

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Performance Summary

Feature	Outcome
Resume Parsing	Accurate extraction of key fields
Job Parsing	Effective keyword identification
Resume Tailoring	Improved alignment with job roles
ATS Feedback	Useful insights for improvement
Iteration	Progressive enhancement of resume

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7. Conclusion

This project demonstrates the effective use of Large Language Models for automating resume creation and optimization. By integrating resume parsing, job description analysis, and iterative refinement, the system provides a comprehensive solution for generating ATS-friendly resumes.

Future Enhancements

- Integration with advanced LLMs for higher accuracy
- Addition of professional resume templates
- Integration with LinkedIn profile parsing
- Advanced ATS scoring algorithm